City Sense: Personal Sensing for Urban Spaces

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Abstract

With practitioners working towards the possibility of truly 'smart' cities, there is a prevalent ideology that smartness has to be external. However, the urban space is one that embodies smartness in and of itself. In this regard, we propose the City Sense framework. A framework that leverages personal sensing for urban sensing via passive crowd-sourcing. To illustrate, the medical wearables market was estimated to be worth 4 billion dollars in 2017, with an expected increase of 35% in 2019. These devices are able to classify our affective states using information gathered from our physiological responses to stimuli. Furthermore, crowd sourcing has proved to be an effective means of harnessing knowledge cheaply but accurately. What we propose, is a system that semantically enriches the information we have of our living spaces using perceptory informatIon gathered from residents of the city. For a long time, scientists in neuro-architecture and cognitive geography have discovered a correlation between the geometry and peculiarities of areas and the moods and psychology of its inhabitants. This framework will be beneficial towards catering for special citizens such as the autistic, the old among others. Additionally, it will inform our design of cities for the future.

Problem

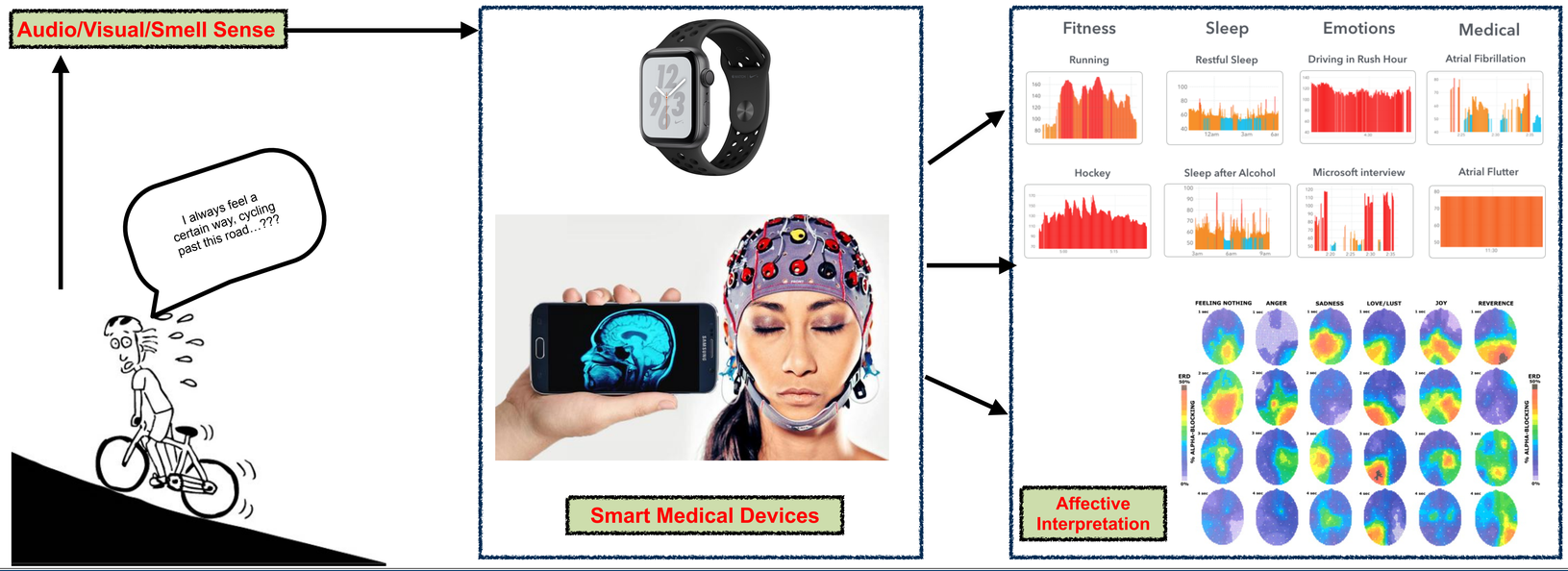
- Our cities are represented using maps that are powered by algorithms with metrics such as shortest path. Whereas a semantically enriched map would provide an even better experience for city dwellers.
- Over 67% of the world's population are projected to reside in urban centres by 2020. This suggests that we have to be proactive in proffering innovative approaches to accommodate this growth.

  "We shape our cities and afterwards our cities shape us" - Anonymous
- The urban land scape is one that is complex in its morphology and evolution. To understand and leverage the complexities that our urban spaces provide. We need to treat smartness as an embodiment of the urban system.

  "Smartness is embodied in the city" - Marcus et al 2017
- Studies have found that you have a 55% chances of suffering from anxiety or depression just by living in the city. This tells a lot about the future of the urban living what with the expected growth. We need to understand the bijective relationship between us and the city.

  "Our daily urban experiences are the product of our perceptions and senses" - Quercia et al 2014

Motivation



Approach

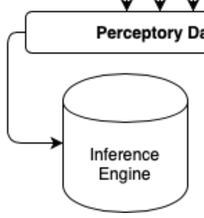
Perceptory data is collected from residents of the city with respect to space. Their physiological response to environmental stimuli is used to develop an affective inference engine. This engine is able to map affect to space and maps this spaces to individual profiles based on health, mental or psychological similarities.

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The semantic enrichment of the city is done via the affective inference engine. Not only are we able to classify spaces into affective categorisations. We are also able to personalise the affective experience of the urban dweller by making spatio-temporal recommendations based on a profile that has been developed.

Objectives, Questions & Applications

Our objectives are to:

- ☑ **Build a perceptory database of the urban space. Using information collected from residents who make use of those spaces.**
- ☑ **Develop a mapping of the physiological responses of users to affective states.**
- ☑ **Use the information of states to enrich the semantic knowledge of spaces.**
- ☑ **Build an affective profile and recommend spaces to users based on affective similarity and need.**

Thus, we ask the following questions:

- ☑ **Can we build an affective profile of residents based on their physiological response to stimuli?**
- ☑ **What are the efficient ways to semantically enrich urban spaces using knowledge derived from their physiological responses?**
- ☑ **How do we accurately evaluate the semantic enrichment models (spatio-temporal)?**

It is a fact that the layout of our buildings and streets has some effect on our mental state and emotional development (The Pruitt-Igoe Myth). Total remodelling is infeasible but we could be able to work around this by enriching our understanding of living spaces using our physiological responses to urban stimulus. If we were to take advantage of the complex nature of our urban space. We will be able to in the least, have answers for certain urban phenomena such as the high risk of suffering mental health problems just by living in the city.

The applications of our framework include:

- ☐ **Recommending places or venues to residents based on their affective similarities or need e.g places that could lighten their moods based on the average affective state of past visitors.**
- ☐ **Recommending places to people with a special need say chronic anxiety against visiting certain areas that are notorious for spiking anxiety in individuals. e.g autistic kids, older citizens.**
- ☐ **Understanding the interaction between layers of the spatial system and how it influences the affective characterisations of areas.**

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